

GridPowerAgent: A Seeded Corpus and Observation-Ablation Benchmark for LLM Power-System Event Diagnosis

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Abstract—Large language models (LLMs) could act as a coordination layer between grid information, analysis tools, and operators—provided they read operating states accurately, select appropriate tools, and avoid hallucinated advice. We present GridPowerAgent, an agent that observes simulated power-system states, retrieves operating procedures, and orchestrates power-flow, contingency, and optimal-power-flow tools. The evaluation is a paired 140-scenario pilot per configuration on IEEE-14—with a 140-scenario IEEE-39 draw rescored under the revised key—run on a seeded, fully regenerable corpus of 16,000 operating points and 15,000 scenarios across the IEEE 14/39/118 systems (9.54 million nested measurements), with power-flow ground truth, noisy measurements, state estimates, and rule-based labels for ten disturbance classes. The pilot exposed and repaired a flaw in the benchmark itself: on the original key, both models scored zero on the two outcome-labeled classes (E6 undervoltage, E8 thermal overload) while describing the injecting cause correctly in nearly every failed response; the corpus now keys every scenario to its injected mechanism. Because diagnosis with the event description in the prompt is trivially leaked (100%), the revised benchmark evaluates diagnosis from observations only. The finding is stark: from EMS-grade telemetry the API model reaches 93.9% and a gradient-boosted classifier on identical features 100%; from raw state the LLM collapses to 8.0% where the classifier reaches 64.3%; the local model manages 43.8% from telemetry (20-scenario probe, $p=9.1e-13$ vs. API). A closed-loop extension in which the agent must identify the switched element by name and chain at least two real tool executions yields 30.8% (API). Tool selection retains a style difference (the API model defaults to power flow in 57% of picks). All results are exploratory single temperature-zero runs with exact denominators, at pilot scale.

Index Terms—power system operation, large language models, LLM agent, tool orchestration, retrieval-augmented generation, state estimation

I. INTRODUCTION

Power-system operators are surrounded by analytical tools—power flow, state estimation, contingency analysis, optimal power flow—yet the work of interpreting their outputs, relating them to operating procedures, and deciding what to do next remains manual and expertise-bound. Large language models (LLMs) are candidates for an *intelligent coordination layer* in this workflow: not replacing conventional analysis, but reading grid states, retrieving the right procedures, invoking the right tools, and explaining the results [1], [2]. Recent agentic systems move in this direction [3], [4], but published evaluations rarely state what the labels are, where they come from, or how the scoring could be reproduced.

This paper presents GridPowerAgent and its evaluation under six research questions: (RQ1) how accurately can an LLM identify normal, abnormal, and critical operating conditions; (RQ2) can it determine which engineering tool a given event requires; (RQ3) can it construct valid tool calls; (RQ4) does grid-specific retrieval improve operational reasoning; (RQ5) does tool grounding reduce hallucination; and (RQ6) does the evaluation transfer across network scale. RQ3—constructing valid tool calls—is answered here at its execution level: stated tools are executed against the scenario’s post-event network after the run and their execution scored (Sec. VII-F). This pilot answers RQ1, RQ2, RQ4, RQ5, and RQ6, with RQ3 answered by a closed-loop extension in which the agent’s tool calls are constructed, executed, and interpreted in-loop (Sec. VII-F). The paper’s identity is deliberately that of a benchmark-and-evaluation study; its contributions are the corpus with dual-axis keying, the observation-ablation methodology, the tier discrimination under realistic observations, and the closed-loop validity protocol—not a new algorithm. The specific findings: observation encoding, not model choice, decides what is diagnosable (telemetry 100%-separable for a classifier, raw state not); description-reading saturates both tiers and conceals their difference; and closed-loop tool use with mandatory evidence-gathering is the only evaluated condition in which the benchmark is neither leaked nor saturated. We answer them on a seeded, headless-resumable corpus pipeline: 16,000 operating points and 15,000 scenarios across IEEE 14/39/118—ten disturbance classes (E0–E9) each carrying pre/post power-flow truth, noisy measurements, state estimates, severity, and rule-based reference labels—plus a FAISS knowledge base of operating procedures and four validated physics tools; configurations (E1 LLM-only, E2 +RAG, E3 +Tools, E4 Full) instantiate the ablation.

The evaluation compares two deployment tiers under identical paired prompts—a 4-bit-quantized small model served locally, and a lightweight API model—so conclusions do not depend on one provider. All figures carry exact denominators and Wilson CIs; power is stated alongside every null result; and one finding receives particular attention: both models systematically fail on two outcome-labeled disturbance classes, for a reason traced to the label taxonomy itself rather than to either model.

Contributions: (i) a grid-aware agent that couples grid-

state observation, procedure retrieval, and validated physics tools; (ii) a seeded, hashed, fully regenerable 16k/15k/9.5M-measurement corpus with rule-based tool supervision and a quantified label-noise bound under estimation uncertainty; (iii) a dual-definition tool-scoring protocol that exposes and removes a degenerate answer strategy; (iv) a paired, exactly-denominated pilot across two deployment tiers in which a mixed-axis labeling flaw was identified and repaired—flipping diagnosis parity into a significant tier gap—with an observation ablation separating description-reading from telemetry-based diagnosis; and (v) a per-check validation disclosure—including unresolved IEEE-39 islanding cases with their scenario identifiers shipped alongside the corpus—together with a seeded IEEE-39 replication of the full paired protocol (Sec. VII-D).

II. RELATED WORK

LLMs have been explored for power-system analysis assistance, dispatch, and contingency response [1], [2], including agentic orchestrators [3], [4] and dispatch benchmarks [5], but published evaluations seldom disclose how reference labels are produced; our corpus couples per-scenario power-flow truth, measurements, state estimates, and rule-based tool supervision across three IEEE sizes under one seeded pipeline, scored with exact denominators. Foundational RAG [6], tool-augmented LLMs [7], [8], [9], classical state estimation [10], [11], contingency/OPF analysis [12], [13], [14], ECE calibration [15], and hallucination taxonomies [16] supply the component methods. Synthetic scenarios inherit the spirit of grid-ML benchmarks [17], [18]; pandapower [19] supplies AC power flow; FAISS [20] with sentence-transformers [21] supplies retrieval. On quantization: 4-bit precision is near-optimal [22], [23], though it degrades small models—a conservative bias in our local deployment (Sec. VIII).

III. METHODOLOGY

Fig. 1 overviews the pipeline. Stages 03–05 build networks, operating points, and scenarios; 06–09 synthesize measurements, state estimates, severity, and reference labels; 10–14 expose physics tools; 16–17 build the knowledge base; 19–22 run the agent ablations; 23–28 evaluate and render figures. Heavy stages are idempotent and headless-resumable, materializing the full corpus in ~ 30 min.

Agent workflow. The agent (Fig. 2) follows an observe–diagnose–retrieve–plan–execute–interpret loop: it receives the post-event structured grid state (voltage magnitudes, branch loadings, outages, storage state of charge); optionally receives the top- k retrieved operating procedures (E2/E4) and a tool manifest (E3/E4); and must return the disturbance class, a tool selection, and a recommendation with confidence, as JSON. The harness scores the response against the rule-based reference labels; it does not execute tools inside the agent loop in this pilot—stated tools are executed post hoc for an execution-validity check (Sec. VII-F). The diagnosis prompt presents the revised cause-axis taxonomy (Sec. VII-A), and

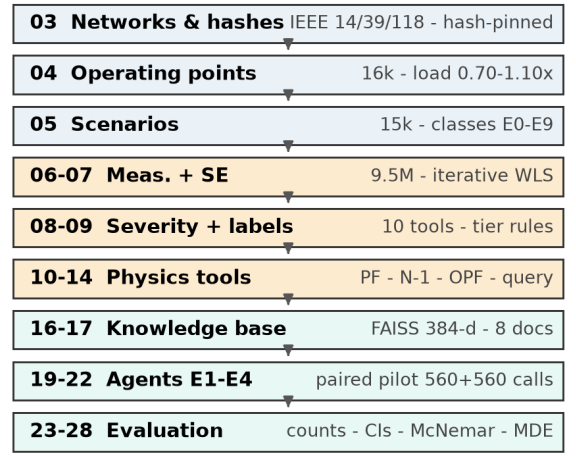


Fig. 1. Pipeline (Stages 03–28).

TABLE I
CORPUS. VALIDATION IS REPORTED PER SYSTEM, NOT AS A SINGLE GLOBAL CERTIFICATE.

Case	OPs	Scen. (per class)	Meas.	Auto checks	Worst V
14 (ref)	4k	3k (300)	361k	21/21	0.89 pu
39	5k	5k (500)	1.50M	20/21	0.89 pu
118	7k	7k (700)	7.68M	21/21	0.89 pu
Total	16k	15k	9.54M	—	—

IEEE-14 limits are artificial (3%/4%) to expose E8. The single failing check (s5_no_nan on IEEE-39) covers 41 scenarios with NaN post-voltages from two islanding outages; identifiers and causes ship with the corpus (case39_nan_scenarios.csv).

three observation conditions (description, telemetry, state-only; Secs. VI and VII-C) probe what the diagnosis measures.

Networks and scenarios. Networks use pandapower IEEE 14/39/118 cases with tuned thermal limits to make compound events observable (14: 3% line / 4% transformer; 118: 6%; 39: nameplate). Limits are disclosed per system and deliberately *not* comparable across systems (Sec. VIII). Operating points sweep load 0.70–1.10 $\times$ with $\pm 5\%$ bus-local noise, renewable fractions, and storage state of charge 0.15–0.85 (20 MW/40 MWh at bus 9 on IEEE-14). Topology hashes pin all three networks. Ten classes are injected: E0 Normal; E1 load surge, E2 load drop, E3 line outage, E4 generator outage, E5 renewable ramp (cause classes); E6 undervoltage, E7 overvoltage, E8 thermal overload (outcome classes, physically iterated to target severity); E9 compound. Reference labels key every scenario to its injected mechanism class; the original outcome-class keying of E6/E8, the flaw it created, and its repair are analyzed in Sec. VII-A.

IV. CORPUS, SIMULATION AND VALIDATION

Table I summarizes the corpus. Scenarios inject the ten classes with 300/500/700 per class on 14/39/118; replay over 40 draws confirms exact determinism.

Measurements model $\sigma_V = 0.003$ pu and $\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05)$ MVA computed from the noise-free power-flow solution. Each scenario carries ~ 120 noisy measurements (bus-voltage magnitudes, branch P/Q at both

TABLE II
DIAGNOSIS UNDER THE REVISED TAXONOMY: EXACT COUNTS, WILSON
95% CIs (PERCENTAGES).

Cfg	API ($n=140$)	Local ($n=140$)
E1	140/140 [97, 100]	137/140 [94, 99]
E2	140/140 [97, 100]	133/140 [90, 98]
E3	140/140 [97, 100]	138/140 [95, 100]
E4	140/140 [97, 100]	134/140 [91, 98]

unit), and single temperature-zero runs capture neither decoding nor prompt variability.

VII. RESULTS

A. A Mixed-Axis Labeling Flaw and Its Repair

The original corpus keying assigned the two *outcome* classes—E6 undervoltage and E8 thermal overload—to scenarios injected through *cause* mechanisms: outages, load surges, and outage-plus-surge compounds. Under that key, both models score zero in every configuration: 0 of 36 pooled E6 and 0 of 26 pooled E8 observations. The failures are systematic: across all 144 failed E6 responses, 128 predict E3, 16 predict E4; across all 104 failed E8 responses, 101 predict E9, 3 predict E3 (“...a single transmission line outage (line_6_13), which directly corresponds to class E3”). The mechanism is a mixed-axis label taxonomy: both models classify on the cause axis and are *causally correct but label-wrong* on every E6/E8 scenario—a labeling-scheme property, not a model property.

The repair rekeys every E6/E8 scenario to its injected mechanism (line outages to E3, generator outages to E4, compounds to E9, surges to E1), re-derives tool tiers, and presents the cause-axis taxonomy to the agent; scenario identifiers, physics, and outcome flags are unchanged (E7 is kept: its mechanisms match the outcome reading). The revised-key pilot uses the same 140 scenarios, prompts, and scoring pipeline.

B. Re-Measured Diagnosis on the Revised Taxonomy (RQ1/RQ4)

Table II reports exact counts under the revised key, with the event description present in the prompt. The API model diagnoses 140/140 in *every* configuration; the local model diagnoses 133–138/140 (95.0–98.6%). This condition is leaked—the description states the cause in plain language (Sec. VII-C)—so these numbers measure reading, not grid reasoning. All 18 vs 0 paired discordances favor the API model (pooled exact McNemar $p=7.6e-06$; Table III, Fig. 3; individually significant in E2 and E4)—the statistical parity observed under the flawed key (77.9% vs. 76.2%, exact McNemar not significant) was an artifact of the mixed axis, not evidence of equivalent capability. Configuration additions move local diagnosis by at most five scenarios in either direction while the API tier stays at ceiling: no diagnosis benefit from retrieval or tool manifests is detectable at this scale. Because the description already carries the diagnosis, this design cannot distinguish RAG inefficacy from information sufficiency; hard-negative retrieval and faithfulness evaluation are required before any conclusion about retrieval quality.

TABLE III
PAIRED DIAGNOSIS ANALYSIS (LOCAL — API, PP), REVISED TAXONOMY:
BOOTSTRAP 95% CI, 20K RESAMPLES; ALL DISCORDANCES
ONE-DIRECTIONAL.

Cfg	Pairs	Diff (pp)	95% CI	McNemar p
E1	140	-2.1	[-5.0, 0.0]	0.25
E2	140	-5.0	[-8.6, -1.4]	0.02
E3	140	-1.4	[-3.6, 0.0]	0.50
E4	140	-4.3	[-7.9, -1.4]	0.03

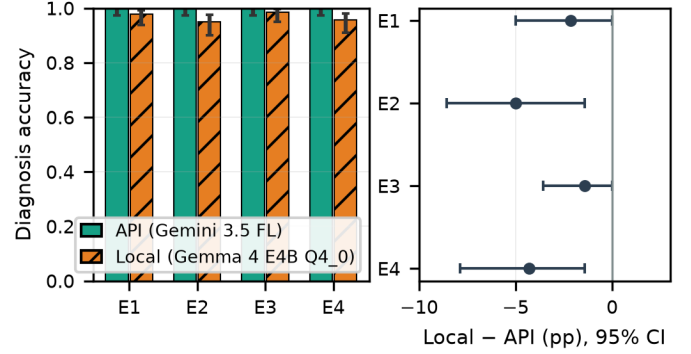


Fig. 3. (a) Diagnosis accuracy under the revised taxonomy, local vs. API model, Wilson 95% CIs. (b) Paired diagnosis difference (Local — API) with bootstrap 95% CIs.

C. Observation Ablation: Reading the Log vs. Reading the Grid

The near-ceiling diagnosis above is measured with a plain-language event description in the prompt. Three observation conditions (Sec. VI) separate what diagnosis measures (Table IV; API tier unless noted). First, the telemetry condition is nearly sufficient for the API model (93.9%; the switch-detectable classes E3/E4/E7 at 100%, compounds 94.1%)—yet a conventional gradient-boosted classifier on the same features reaches 100.0% so the observation is fully class-separable and the LLM’s residual errors are composition errors, not information limits. Second, the raw state condition collapses to 8.0% for the LLM while the same classifier reaches 64.3% from identical features: the signal exists, but the LLM cannot exploit unlabeled numerical summaries, answering E7 for 95% of scenarios and calling every E0 normal scenario E7 under telemetry (0/20). The description-only condition quantifies the textual leak: 100.0% with no grid-state numbers. Multi-axis scoring on pipeline-consistent annotations (rule-generated by the same pipeline as the events—the recursion is disclosed; expert re-annotation remains the controlling gap) shows answers outcome-consistent in 94.3%/93.6% and tool-consistent in 60.2%/59.6%: the revised key no longer punishes cause-correct answers on outcome axes. The telemetry condition also separates the tiers: a 20-scenario local probe reaches 43.8% vs. 95.0% for the API tier (41 vs. 0 discordant, $p=9.1e-13$); its errors include hedged multi-class answers and JSON contract breaks. Replacing the true post-event state with the WLS-estimated state leaves diagnosis at 97.1% over all 560

TABLE IV

DIAGNOSIS BY OBSERVATION CONDITION (API TIER; LOCAL TELEMETRY $n=80$ CELLS). CLASSIFIERS FITTED ON NON-PILOT SCENARIOS, EVALUATED ON THE PILOT DRAW; DESCRIPTION-ONLY IS TEXTUAL (–); WLS ROW USES ITERATIVE-ESTIMATED STATE (SEC. IV), LLM-ONLY.

Condition	API LLM	Local LLM	GBM	Log. reg.	maj./rand.
Event description (leakage control)	100.0	–	–	–	–
Telemetry only	93.9	43.8	100.0	100.0	–
Telemetry, WLS-estimated state ($n=560$)	97.1	–	–	–	–
Raw state only	8.0	–	64.3	47.9	20.0/12.5

TABLE V

TOOL SELECTION, STRICT-SPECIFIC METRIC: EXACT COUNTS, WILSON 95% CIs (PERCENTAGES).

Cfg	API ($n=140$)	Local ($n=140$)
E1	58/140 [34, 50]	81/140 [50, 66]
E2	56/140 [32, 48]	77/140 [47, 63]
E3	49/140 [28, 43]	75/140 [45, 62]
E4	43/140 [24, 39]	82/140 [50, 66]

cells: the tier and observation findings survive realistic state estimation. Estimation even helps slightly (97.1% vs. 93.9%): smoothing pulls marginal violations toward nominal, which the flag-agreement rate (94%)

D. RQ6: Cross-Network Evaluation on IEEE-39

A seeded 140-scenario draw from the IEEE-39 corpus (excluding the 41 islanding-NaN scenarios) runs the same paired protocol under the pre-revision prompt; its logged answers are rescored against the revised key. The API model diagnoses 140/140 scenarios in *every* configuration (E1–E4) and the local model 135–138/140. Strict tool selection declines monotonically for the API model—57, 52, 43, and 31 of 140 across E1–E4—while the local model stays flat at 62–63/140. The decline extends the IEEE-14 pattern (58, 56, 49, and 43 of 140 there) but is steeper and does not plateau by E4, while the local model is flat on both corpora: the API decline across configurations is monotone on both corpora and currently unexplained—flagged for the powered sweep. Transferring between corpora changes the network topology, load patterns, and label instances while the agent, prompt, and scoring pipeline remain untouched.

E. RQ2: Tool Selection and the Default-Answer Bias

Tool selection is where the deployment tiers differ most—in *style* before *accuracy*. The permissive accepted-set metric is degenerate (always answering power flow achieves it by construction; random choice scores 53%), and the API model exercises exactly that strategy: 57% of its stated tools are power flow. Under the strict-specific metric (Table V), the local model names a scenario-*required* tool in 75/140–82/140 of cases (flat across configurations) versus 43/140–58/140 for the API model. When the API model states power flow (317 responses), it is actually required in only 41% of cases—the rest are defaults to the universally accepted answer. The local model states power flow rarely (73 responses, 26% of them required); its dominant choice is contingency (315 responses, 100% required, precisely on outage/overload scenarios). The

gap therefore reflects the API model’s default rate, not superior local tool knowledge; both models remain far from ceiling, and the binary metric cannot distinguish well-placed tools from lucky guesses.

F. RQ3: Closed-Loop Tool Use

The one-shot pilot only *names* tools; RQ3 asks whether the agent can *use* them. The closed-loop extension removes the remaining shortcut: the per-element status table does not say which element was switched, at least two real tool executions must precede any answer, and the answer must *name the switched element*—scored against the truly switched element. On 107 pilot scenarios (API tier): diagnosis 30.8%, element construction 71.9% on the 64 switched-element scenarios, over 219 tool calls in 3.59 turns on average (100% of scenarios chain two or more calls). The task is now genuinely hard: construction succeeds while exact-class diagnosis does not follow automatically, and the offline plumbing check (537 stated-tool pairs across the pilot rounds, 521 executable, all succeeding) confirms the failures are the agent’s, not the tools’. What remains open is decision quality—whether recommendations drawn from executed results are operationally safe and constraint-relieving—and the loop is API-tier only; extending it (and the estimated-state arms) to the local tier is future work.

G. RQ5: Hallucination and the Cost of Local Inference

Any-tag hallucinated rows are rare for both models under the automated judge: 0/0/0/0 (E1–E4) (API) and 2/0/0/1 (E1–E4) (Local) across E1–E4. Judge flags are not expert-annotated ground truth; confidence calibration is unidentifiable at this sample size and will be reported on the powered sweep. Latency is the deployment trade-off: local inference averages 44–49 s per call versus 1.06–1.11 s for the API round-trip—roughly a 44 $\times$ difference with zero marginal API cost; acceptable for always-on monitoring, not for closed-loop use (Sec. VIII).

VIII. DISCUSSION AND LIMITATIONS

Scope. The evidence covers *one model pair* (a 4-bit local model; a lightweight API tier), one prompt template, temperature 0, single runs; it does not speak to frontier API models. The powered 600-scenario sweep is the inferential step.

Label circularity. Disturbances, labels, and prompts derive from the same rule family: the evaluation measures *recovery of the synthetic labeling policy*, not open-ended operator reasoning—and the severity boundaries of Eq. (1) were themselves tuned on this corpus, though the strict-metric comparison is invariant across four boundary sets, two of them not derived from this corpus (Sec. IV). Required remedies, all future work: expert-reviewed labels (2×80 , $\kappa \geq 0.8$), multiple valid tool sequences, blind scoring, an evaluator independent of the label generator, and externally anchored severity boundaries.

Taxonomy design. The E6/E8 label-axis failure (Sec. VII-A) was a benchmark-design finding: outcome-labeled classes injected via cause mechanisms made

cause-correct answers score zero. The revision rekeys E6/E8 to their injected mechanisms; both label versions ship with the corpus. The observation ablation adds a second encoding finding: the state-only view carries essentially no cause information (8.0% diagnosis, Sec. VII-C), and the API tier misreads every E0 normal scenario as overvoltage under telemetry-only observation.

Physics fidelity. Thermal limits differ per system by construction, so overload magnitudes are not operationally comparable across systems. The corpus is static: no dynamics, protection, or time-sequential behavior. The severity index is illustrative with a quantified label-noise bound (Sec. IV); the state estimator is an iterative WLS without bad-data handling or topology-error processing (Sec. IV); the 41 islanding-NaN IEEE-39 scenarios (identifiers shipped in `case39_nan_scenarios.csv`) remain unresolved and are excluded from downstream post-voltage use.

Scoring subjectivity. Hallucination flags come from a single automated judge (Sec. VII-G); tool scoring relies on the reference policy; the strict metric narrows the policy-dependence.

Safety framing. This is an offline advisory prototype: no protection coordination, authorization, or human-approval interface is implemented; autonomous control actions are out of scope.

Reproducibility. All scripts, seeds, hashes, prompts, validation summaries, both label versions, and the revised-taxonomy and observation-ablation datasets are in the repository; corpora regenerate deterministically. Generalization: the IEEE-39 evaluation (Sec. VII) covers both tiers; IEEE-118, a powered cross-system protocol, and the powered 600-scenario sweep (per-class F1, ECE, multiplicity-corrected tests, prompt sensitivity) remain future work.

IX. CONCLUSION

GridPowerAgent couples a seeded, regenerable 16k/15k power-system scenario corpus with a grid-aware LLM agent that retrieves procedures and orchestrates validated physics tools, evaluated under a protocol where every number carries an exact denominator. In the paired pilot, a systematic diagnosis failure was traced to the benchmark’s mixed-axis label taxonomy and repaired; on the revised taxonomy the API tier diagnoses at ceiling and significantly above the local tier (100% vs. 96.8%), and telemetry-only diagnosis preserves the gap (93.9% vs. 43.8%). The corpus, protocol, and disclosed flaws are the contribution; the powered evaluation is the next step.

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